Myronas Koulas

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EU citizen (Cyprus, Poland) with UK Settled Status: no sponsorship needed in the EU, Switzerland or the UK · Open to relocating

PROFILE

Control engineer and PhD researcher at the University of Aberdeen. I build AI models of real hardware, then use them to give robots a sense of touch and to train controllers that work on the real machine without retuning.

My origami robot feels pushes without a force sensor, and the same method gave a nanopositioner 952 Hz bandwidth, the highest ever measured on that stage. I work hands-on, from circuit boards to real-time code.

EDUCATION

PhD, Electrical and Electronic Engineering — University of Aberdeen *2025 – 2028 (expected)*

- Robotics and control research in the ARMS Group, in collaboration with EPFL. Supervisors: Dr Andres San-Millan, Prof. Sumeet S. Aphale.

MEng, Electrical and Electronic Engineering, First Class Honours — University of Aberdeen *2020 – 2025*

- Dissertation: control of an origami 3-DoF robot arm. Modules: Control Systems, Machine Learning, Optimisation, Software Engineering.

RESEARCH AND ENGINEERING EXPERIENCE

Doctoral Researcher — ARMS Group, University of Aberdeen *2025 – present*

One method developed in three steps, from a touch-sensing origami robot to precision motion control.

1. Touch without force sensors (origami robot, with EPFL)

- Designed and built a 3-motor origami robot from CAD to a fully instrumented test rig, including its real-time C/C++ firmware (ESP32) and a control interface, first in LabVIEW and later in Python.
- Made its cheap, rigid geared motors behave softly with **no force sensors**: it feels pushes as light as 0.05 Nm, locates a load to within 2 cm, and a 1 kg weight dropped on it bounces **34% less**.
- Programmed it to feel like anything from a **stiff wall to a soft spring or marshmallow**, and to give touch feedback such as a purring-cat vibration, opening uses in haptics and collaborative robotics.

2. Feeling touch during fast motion (AI motor twin)

- Found that standard methods lose the sense of touch once the robot moves fast, so built an AI **digital twin of each motor** (about 3,000 parameters) that predicts how it should move.
- All three twins run live at once on one low-cost ESP32 chip, so the robot feels a push without false alarms **even while it moves fast**.
- Used this to let people teach the robot by touch: it learns a new path when guided by hand or after bumping into an obstacle.

3. Training controllers with the same twin (nanometre-precision stage)

- Turned the method into a controller-training tool: the controller learns through a frozen model of the machine (Python, PyTorch), then exports automatically to C/C++ for ESP32 and for MATLAB/dSPACE.
- Tested it on a nanopositioner, a compliant stage with the hysteresis, delay and resonance that also make origami robots hard to control. It worked on the real machine first time, with **no manual retuning**, running at 20 kHz.
- Made the stage respond faster than ever measured on it: **952 Hz bandwidth, 32% above the best published hardware result** (720 Hz) and 1.33× its first resonance, with nanometre-level accuracy.
- Introduced a verification step that proves the code running on the machine matches the design, which caught builds silently reusing old code.

Electrical Engineer (part-time) — Metron *Sep 2026 – present*

- Diagnose and repair electronic systems for offshore clients down to component level, using oscilloscopes, soldering and component test equipment to find faults and to repair and refurbish circuit boards.

Carnegie Scholar / Research Intern — Carnegie Trust and University of Aberdeen *Summer 2024*

- Won a competitive national scholarship to design and build the origami robot arm and its first control system.

Undergraduate Research Intern, precision positioning — University of Aberdeen *Summer 2023*

- Developed a new controller design method, tested it on real hardware, and published it at an international conference as an undergraduate.

PUBLICATIONS

- M. Koulas, S. S. Aphale, J. Paik and A. San-Millan, "The Stiffness Paradox: Recovering Compliance in Rigidly Actuated Origami Robotics," *IEEE/ASME Transactions on Mechatronics*, under review, 2026.
- A. B. Gutierrez, M. Koulas, S. S. Aphale and A. San-Millan, "Enhancing Nanopositioning Bandwidth via PAVPF and Pole Placement in the Presence of Delay," *Mechatronics*, under review, 2026.
- M. Koulas, A. B. Gutierrez, S. S. Aphale and A. San-Millan, "Butterworth Pattern-Based Integral Resonant Controller Design for Nanopositioning Systems with Internal Delay," *UKACC 14th International Conference on Control*, 2024.
- M. Koulas et al., "One Learned Feedback Law for Damping, Tracking and Hysteresis Compensation in Piezoelectric Nanopositioners," journal paper in preparation.

- M. Koulas et al., “Can Low-Cost Geared Robots Feel a Push While They Move?” *IEEE Robotics and Automation Letters*, in preparation.

SELECTED PROJECTS

- **Offshore wind farm (MEng group project):** designed the power electronics and control for each 8 MW turbine of a 120 MW wind farm in Simulink: generator- and grid-side converter control (field- and voltage-oriented), three-level converters with DC-link balancing, and fault protection, verified against grid-code rules. Led the electrical sub-team in a multidisciplinary team of electrical, mechanical and civil engineers, working with the project lead to keep the design consistent across disciplines.
- **Electronics design:** designed and built a motor-controller board and a fire-detection device from schematic to working prototype.
- **Self-balancing robot:** 1st place in a university-wide competition (three-person team).
- **Vision-based navigation:** maze-solving robot in CoppeliaSim using OpenCV and TensorFlow, trained and tested in simulation.

SKILLS

- **Control and AI:** control system design, system identification, machine learning for control (PyTorch, TensorFlow), digital twins, force and compliance (admittance) control without force sensors, haptics, Kalman filtering, PID.
- **Software and real-time:** Python, C/C++, MATLAB/Simulink, dSPACE, LabVIEW, Git; embedded firmware on ESP32 and Arduino (FreeRTOS, SPI, I^2C).
- **Hardware:** PCB design and repair, soldering, oscilloscopes, sensors and motor drives, power electronics, SolidWorks.

AWARDS, LEADERSHIP AND LANGUAGES

- Carnegie Vacation Scholarship (2024); First Class Honours (2025); 1st place, university robot competition.
- Led two student societies: revived the IEEE Student Branch as Chair, and doubled EEE Society membership as President. Teaching demonstrator in Maths, Programming and Engineering Design. Corporal, Cypriot National Guard, leading a team of 20.
- Languages: English, Greek and Polish (fluent); German (beginner).